

cellqc QC report

Single-cell RNA-seq quality control

2026-08-05

1 sample | cellqc v0.2.0 | random seed 42

What this run did

Pipeline

- 1 Ambient RNA: **soupx**
compared (not applied): **decontx**
- 2 Filter on counts, genes, % mitochondrial
- 3 Doublets: **doubletfinder**, **scdblfinder**
cells removed by **doubletfinder** only
- 4 Nuclear fraction: 1/1 sample

Thresholds

- min UMI per cell: 500
- min genes per cell: 300
- max % mitochondrial: 10
- expected doublet rate: 0.1 at 13000 cells/reaction

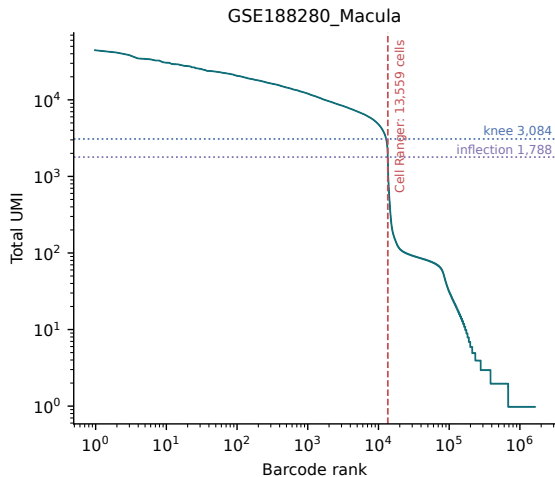
Cells retained at each stage

Sample	Cell Ranger cells	After count filter	Removed (count)	After doublet	Removed (doublet)	Retained
GSE188280_Macula	13559	11234	2325	10223	1011	0.754

Every exclusion is the difference between two adjacent columns.

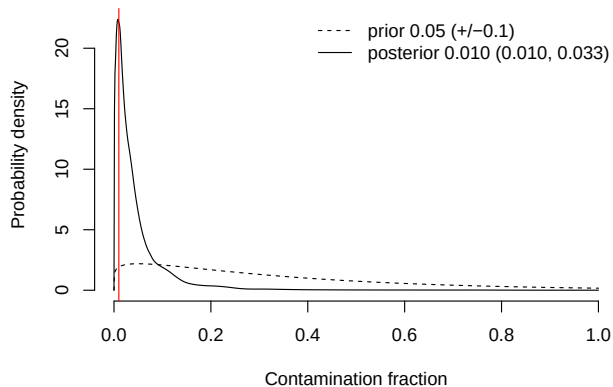
Metric	Value	Metric	Value
Estimated Number of Cells	13,559	Reads Mapped to Genome	96.7%
Mean Reads per Cell	17,203	Reads Mapped Confidently to Genome	93.3%
Median Genes per Cell	2,767	Reads Mapped Confidently to Intergenic Regions	6.3%
Number of Reads	233,256,198	Reads Mapped Confidently to Intronic Regions	53.2%
Valid Barcodes	97.3%	Reads Mapped Confidently to Exonic Regions	33.8%
Valid UMI Sequences	99.9%	Reads Mapped Confidently to Transcriptome	63.0%
Sequencing Saturation	28.1%	Reads Mapped Antisense to Gene	23.4%
Q30 Bases in Barcode	95.9%	Fraction Reads in Cells	90.6%
Q30 Bases in RNA Read	93.6%	Total Genes Detected	33,530
Q30 Bases in UMI	95.7%	Median UMI Counts per Cell	6,163

GSE188280_Macula: Barcode rank



Cell Ranger EmptyDrops call (dashed line) against the UMI-rank curve. A clean library shows the call near the knee. Diagnostic only — cellqc does not re-call cells.

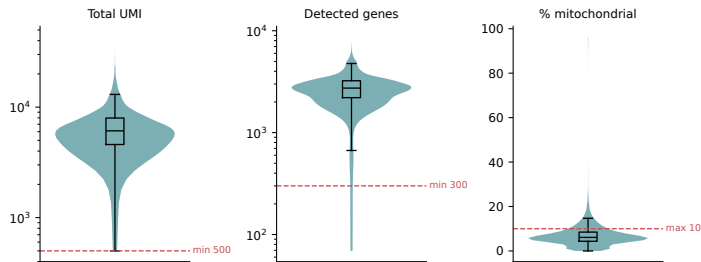
GSE188280_Macula: SoupX rho = 0.010



Estimated contamination fraction. This correction WAS applied to the counts.

GSE188280_Macula: QC metrics before filtering

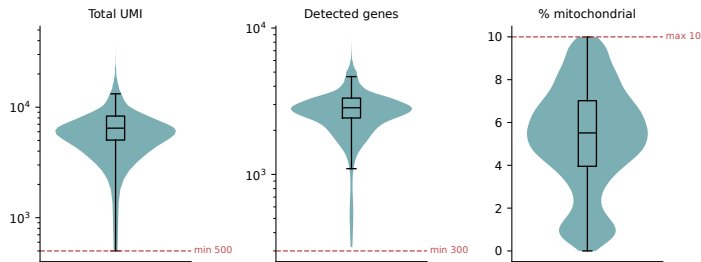
GSE188280_Macula — before filtering (n=13559)



Dashed lines are the applied thresholds.
Counts and genes on a log scale.

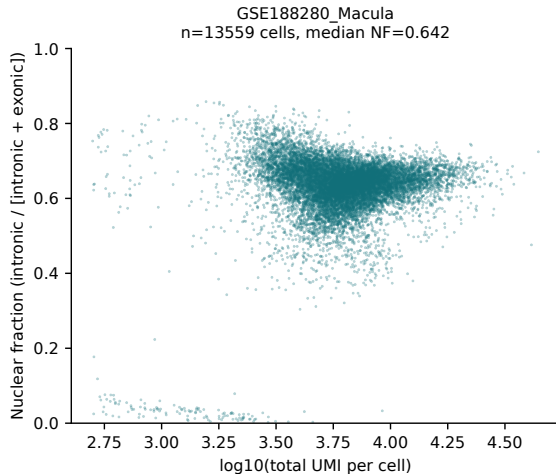
GSE188280_Macula: QC metrics after filtering

GSE188280_Macula — after filtering (n=11234)

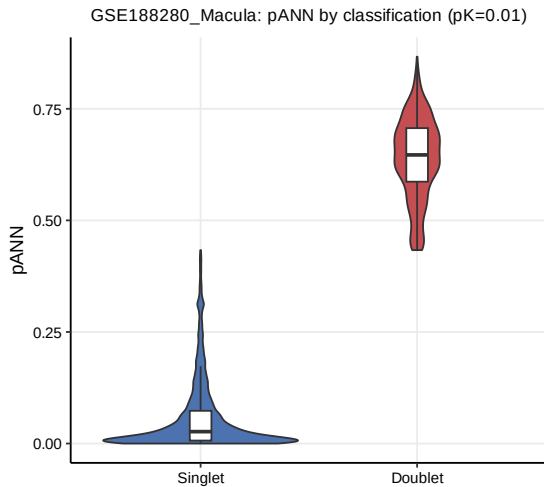


Same axes as the previous slide, after removing cells failing any threshold.

GSE188280_Macula: Nuclear fraction vs UMI depth

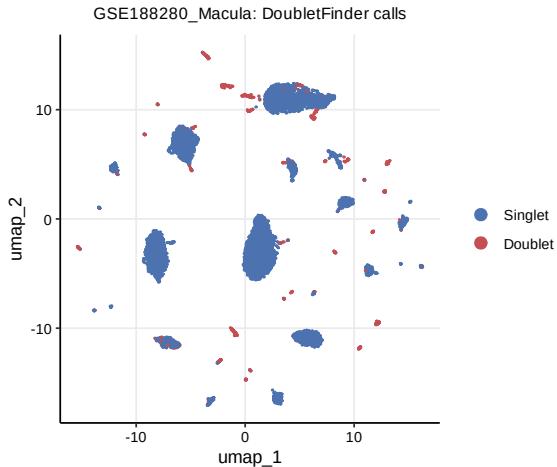


Intronic / (intronic + exonic) reads per cell against sequencing depth. High nuclear fraction at low depth suggests damaged cells or free nuclei. Reported only — NOT used for filtering.

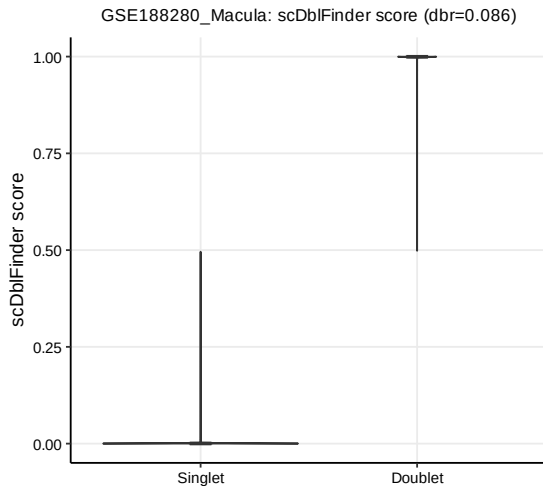


Proportion of artificial nearest neighbours,
by call.

GSE188280_Macula: DoubletFinder: calls on UMAP



Doublets should sit between transcriptionally distinct populations, not form their own island.



Run with the same expected doublet rate as DoubletFinder so the two are compared under matched assumptions. Diagnostic only.

Limitations to carry with these results

- Homotypic doublets are not modelled (`modelHomotypic` is deliberately not called), so the expected-doublet count over-estimates the DETECTABLE doublet count and the doublet step removes slightly more cells than the true heterotypic count. The bias direction is known and constant.
- The expected doublet rate is the 10x rule of thumb ($\text{rate}=0.1$, $\text{capacity}=13000$ cells per reaction), not a measurement for this library.
- Cell calling is Cell Ranger EmptyDrops. `cellqc` does not re-call cells; the barcode rank plot is diagnostic only.
- The nuclear fraction is reported but NOT used for filtering. DropletQC-style empty-drop and damaged-cell thresholds are sample- and tissue-dependent, so applying them automatically across a cohort would be unreviewed auto-filtering.
- Two doublet callers ran; only `doubletfinder` removed cells. The concordance table is a consistency measure, NOT evidence that either caller is correct – with no ground-truth doublets neither can be shown superior on this data.
- Ambient correction applied: `soupx`. Also estimated for comparison: `decontx` (these did NOT modify counts).
- Random seed 42 was set for every stochastic step. `v0.1.0` set no seed and its doublet calls were not reproducible.